Artificial intelligence

Lec (I)

Lecture (1)

HISTORY OF AI

- Development of **artificial neurons** → laying the groundwork for neural networks
- Alan Turing → proposes the **Turing Machine** & the **Turing Test**
- Birth of AI → in Dartmouth Conference
- ELIZA
  - First chatbot
  - simulating human conversation
- WABOT-1
  - First intelligent robot
- First AI winter → a period of reduced funding and interest in AI research
- Expert systems
  - AI programs
  - mimic human expertise
- Second AI winter
- IBM's Deep Blue
  - The first computer to defeat a world chess champion
- Roomba
  - AI enters homes
  - Is an autonomous vacuum cleaner
- IBM's Watson
  - wins the quiz show *Jeopardy*
- Google Now
  - AI-powered personalized assistance
- Chatbot Eugene Goostman
  - passes Turing Test
- Amazon Echo
  - voice-activated AI assistants

TURING TEST

- designed to provide a satisfactory operational definition of intelligence
- **A computer passes the test if**
  - if a human interrogator, after posing some written questions
  - cannot tell whether the written responses come from a person or from a computer
- The computer would need to possess the following capabilities:
  - Natural language processing
    - to enable it to communicate
  - Knowledge representation
    - to store what it knows or hears
  - Automated reasoning
    - to use the stored information
      - to answer questions
      - to draw new conclusions
  - Machine learning
    - to adapt to new circumstances
    - to detect & extrapolate patterns
  - Computer vision
    - to recognize objects, faces, and scenes
  - Robotics
    - to manipulate objects
    - navigating environments
    - perform physical tasks

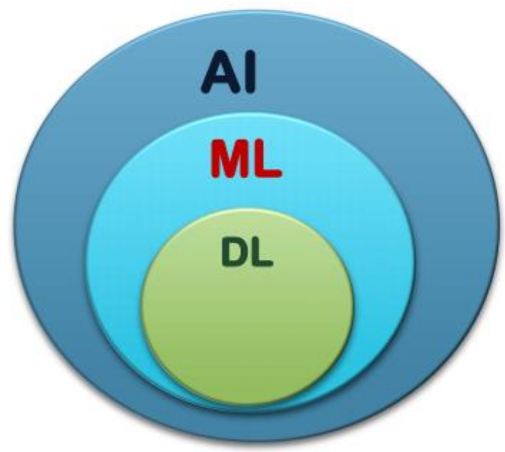
AGENT

- Agent
  - something that acts
  - Software or Hardware
- Rational agent
  - acts to achieve the best outcome
  - OR the best expected outcome → if there is uncertainty
- Intelligence
  - concerned mainly with rational action
  - intelligent agent → takes the best possible action in a situation

AI

- Different people approach AI with different goals
- Think
  - think like humans
    OR
  - think rationally
- Act
  - act like humans
    OR
  - act rationally

- AI
  - Enable computers to mimic human behavior
- ML
  - The ability to learn from the data
  - without being explicitly programmed
- DL
  - Deep layers of neural networks
  - utilized to extract features (pattern) from the data



ما أجمل أن تتوافق أحلامك مع أقدار الخالق عزوجل...